

EXPERIMENT NO. 18:

Implement Perceptron based IRIS Classification

Aim:

To implement Iris Flower Classification using the Perceptron algorithm in Python and classify Iris flower samples.

Procedure:

- 1. Import the required Python libraries.**
- 2. Load the Iris dataset.**
- 3. Separate the input features and target classes.**
- 4. Split the dataset into training and testing sets.**
- 5. Create a Perceptron classifier.**
- 6. Train the model using the training dataset.**
- 7. Predict the class labels for the test data.**
- 8. Calculate and display the classification accuracy.**
- 9. Predict the species of a new Iris flower sample.**

Program:

```
# Iris Classification using Perceptron  
  
from sklearn.datasets import load_iris  
  
from sklearn.model_selection import train_test_split  
  
from sklearn.linear_model import Perceptron  
  
from sklearn.metrics import accuracy_score  
  
# Load Iris Dataset  
  
iris = load_iris()  
  
X = iris.data  
  
y = iris.target
```

```
# Split dataset

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Create Perceptron Classifier

model = Perceptron(max_iter=1000, random_state=42)

# Train the model

model.fit(X_train, y_train)

# Predict test data

y_pred = model.predict(X_test)

# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:")

print(round(accuracy * 100, 2), "%")

# Predict a new Iris flower

new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("\nPredicted Iris Flower:")

print(iris.target_names[prediction][0])
```

Output:

```
... Accuracy:
46.67 %

Predicted Iris Flower:
setosa
```

Result:

The Perceptron-based Iris Classification was successfully implemented in Python. The model classified the new Iris flower sample as Setosa and achieved high classification accuracy.

